

RATISH PATIL

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- [LinkedIn](#)
- [Github](#)
- [Portfolio](#)

Education

SIES Graduate School of Technology

B.E. Artificial Intelligence and Data Science

Nerul, Navi Mumbai

2023 – 2027

- CGPA: 9.26
- **Relevant Coursework:** Data Science, Data Structures and Algorithms, Machine Learning, Data Warehousing and Mining, Computer Vision, Web Development

B.K. Birla College (HSC)

Higher Secondary Certificate (Science)

Kalyan, Maharashtra

2021 – 2023

Academic & Personal Projects

KrishiSetu: AI Vernacular Assistant for Farmer | [Github](#)

Jan 2026 – Present

Node.js, React, RAG, YOLOv8-cls, FastAPI, Gemini API, Tesseract OCR, HuggingFace, Google News API, Docker

Nerul, Navi Mumbai

- Computer Vision Pipeline: Trained a custom YOLOv8 classification model on 22K+ images to diagnose 24 distinct crop diseases, achieving 98% accuracy while optimizing the architecture for efficient, lightweight CPU inference.
- Multimodal NLP Voice Assistant: Engineered a multilingual conversational AI agent that enables hands-free platform navigation, allowing users to query complex government schemes and instantly check eligibility via voice commands.
- OCR, Matchmaking & RAG: Architected an automated pipeline using Tesseract OCR to parse unstructured land records (7/12 documents). This feeds into an LLM-driven matchmaker that ranks scheme eligibility and utilizes a RAG workflow to generate personalized, downloadable PDF treatment and health reports.
- Data Ingestion & Offline Sync: Integrated the Google News API to scrape and serve real-time agricultural updates (Taaza Khabar), alongside deployed SMS webhooks to dispatch critical scheme and weather data to farmers in zero-connectivity areas.

Smart Sight: AI Navigation for Visual Impairment | [Github](#)

Aug 2025 – Nov 2025

Raspberry Pi 4B, YOLOv4-tiny, OpenCV, Tesseract OCR, Ollama, Python

Nerul, Navi Mumbai

- Developed a standalone, multimodal Edge-AI system on a Raspberry Pi 4B integrating real-time computer vision, local NLP, and an event-driven cloud synchronization pipeline via the Microsoft OneDrive API.
- Deployed an optimized YOLOv4-tiny and OpenCV pipeline for low-latency obstacle detection alongside Tesseract OCR for real-time text and sign recognition.
- Engineered a fully offline local voice assistant using TinyLlama and faster_whisper to process speech-to-text and conversational intent completely on-device.
- Optimized multi-model inference and network I/O workloads via intermittent frame processing, sustaining a stable 10 FPS while maintaining real-time responsiveness within strict hardware compute limits

Review Insight Navigator (Amazon Sentiment Analyser & Price Tracker) | [Github](#)

Feb 2025 - April 2025

Python, BeautifulSoup, Selenium, Scikit-learn, Streamlit

Nerul, Navi Mumbai

- Built an automated web scraper using BeautifulSoup and Selenium to extract product reviews and price history across **30+ Amazon product listings**.
- Applied NLP-based sentiment classification using Scikit-learn achieving **85% accuracy**, categorizing customer opinions and surfacing key product trends from unstructured review data.
- Visualized price trends and sentiment breakdowns via Matplotlib and deployed an interactive Streamlit dashboard for end-user exploration.

Technical Skills

- **AI, NLP & Computer Vision:** LangChain, LangGraph, RAG, YOLOv8, Pytorch, OpenCV, Scikit-learn
- **Languages & Backend:** Python, SQL, Node.js, FastAPI, Flask, REST APIs
- **Data Pipelines & Web Scraping:** Firecrawl, TinyFish, BeautifulSoup, Selenium
- **Databases & Cloud Services:** PostgreSQL, MongoDB, Supabase, Firebase, Google Colab, AWS, GCP, Docker, Render, Vercel
- **Web Frontend & Version Control:** React.js, Streamlit, Git, GitHub
- **Coursework & Certifications:** Advanced RAG & LLM Architectures (CampusX), [21 Days, 21 Projects — GeeksforGeeks](#): ML/DS Applied Projects

Leadership & Activities

National Service Scheme (NSS)

Nerul, Navi Mumbai

Volunteer

Aug 2024 – Jan 2025

- Co-organized the "DREAM RUN" marathon, managing logistics and coordination for over 1,000 participants. Actively participated in and promoted community-focused events, including Blood Donation Drives and Tree Plantation Drives.
- Performed in and assisted with organizing skits for Independence Day and Republic Day celebrations.